Guided Assignment 3

Hints and worked solutions

Use sparingly. Attempt the student assignment first. This page contains hints only; worked answers start on the next page. After checking, close this file and reconstruct the argument. Correct units, dimensions, domain assumptions, and causal explanations matter more than matching wording.

One hint per stage

1. Normalize the vector before taking its encoder projection. A decoder has one row per output coordinate.

2. Compute each coordinate separately, then check the vector norm. Transforming a decoded vector happens before re-encoding it at the receiver.

3. Substitute $X_R = RX_1$ in the decoder formula. The activity matrix is unchanged only at corresponding normalized positions.

4. Solve for the exponential in the rate expression first. Two current constraints determine gain and bias; preserving those constraints requires recalibration after a time-constant change.

5. The Y recording sees a connection filter as well as its probe. A linear map sends zero to zero; an affine map need not.

6. Expand the filter equation and compare the coefficients of state and input with the desired derivative. Keep milliseconds out of a calculation otherwise using seconds.

7. Pulse area becomes a held state. Integrate the ramp input rather than copying its shape. Evaluate the sine antiderivative at both limits.

8. Rate mode removes individual spikes, but preserves approximate decoding and recurrence. Variable feedback needs the product of the state and the control input, so preserve both in a joint representation.

9. No new technique is needed. If a part stalls, find its earlier stage and redo a nearby exercise before returning.

1. Representation chain

1a. $\mathbf{x}$ is the represented vector; R sets its scale/domain; $\mathbf{e}_i$ is a unit preferred direction; α_i scales normalized projection into current; β_i is the neuron's additive bias current. There are two represented coordinates and 100 currents, one per neuron.

1b. The projection is $\mathbf{e}^T\mathbf{x}/R = (3/5 + 4/5)/2 = 0.7$. Thus $J = 2(0.7) + 0.3 = 1.7$. A current is an input to a neuron's nonlinear response, not a coordinate of $\mathbf{x}$. The population pattern is decoded to estimate coordinates.

1c. $D \in \mathbb{R}^{2\times n}$ and $\hat{X} = DA \in \mathbb{R}^{2\times N}$. Nengo constructs tuning parameters, evaluates activities over a domain, solves for decoders, and combines them with receiving encoders, gains, radius scaling, and transforms to build weights. Finite basis coverage, regularization, neural dynamics, and spiking fluctuations prevent exact general reconstruction. Encoding and decoding are distinct steps.

2. Vector computations

2a. The contributions are $(0.5, 1)$, $(-0.3, -0.9)$, $(0.4, 0.2)$, and $(-0.8, 0.4)$. Therefore $\mathbf{w} = (-0.2, 0.7)^T$. Draw four 2-D sources into one 2-D receiver, with transforms $I_2, -3I_2, 2I_2, -2I_2$. Every source retains both coordinates.

2b. The target becomes

$$\mathbf{w}(t) = (0.9 - 5\sin(4\pi t),\ 0.7)^T.$$

Its period is 0.5 s. The first coordinate ranges from -4.1 to 5.9; hence the maximum norm is $\sqrt{5.9^2 + 0.7^2} = \sqrt{35.30} \approx 5.94$. That is the minimum ideal spherical radius; practical margin can be useful. The constant $\mathbf{x}$ has norm $\sqrt{1.25} \approx 1.118$, so radius 1 does not cover it. A unit disk does not contain every vector whose individual coordinates lie in $[-1, 1]$.

2c. For each branch $W_X = E_W T_X D_X \in \mathbb{R}^{m\times n}$. With corresponding decoded identities for the other populations,

$$J_i = (E_W)_i \left(T_X D_X \mathbf{a}_X + T_Y D_Y \mathbf{a}_Y + T_Z D_Z \mathbf{a}_Z + T_Q D_Q \mathbf{a}_Q\right) + \beta_i.$$

Add the receiver's neuronal bias once. The activities in this equation have already been filtered consistently. Different source neuron counts give different branch weight widths.

2d. Match input waveforms, representation domains/tuning and decoding conventions, and synapse/probe settings and units. Match neuron realizations and solver settings for closer numerical comparisons. The ideal computation is the same; independent finite spiking realizations generally have different instantaneous fluctuations and possibly different static errors. Nengo automates the representation/transformation construction rather than replacing its mathematics.

3. Radius and static accuracy

3a. Physical intercepts are Rz: $0.125, 0.25, 0.5, 1$ respectively. At fixed $r = x/R$, the normalized current $\alpha r + \beta$ and hence steady rate are unchanged when normalized intercepts, maximum rates, neuron model, and time constants remain fixed. At a fixed physical x, the normalized position generally changes.

3b. Linearity in the targets gives

$$D_R = RD_1, \quad \hat{X}_R = R\hat{X}_1, \quad X_R - \hat{X}_R = R(X_1 - \hat{X}_1).$$

Thus $\mathrm{RMSE}(R) = R\,\mathrm{RMSE}(1)$ for positive R under the stated controls. The predictions are

R	0.5	1	2	4
RMSE	0.015	0.03	0.06	0.12
RMSE/R	0.03	0.03	0.03	0.03

This is an exact scaling argument for corresponding samples and a fixed linear regularized solve, not a universal empirical law for every radius experiment.

3c. A fixed physical test interval corresponds to different normalized subintervals at different radii. It therefore samples different portions of the error curve. Keep the neuron realization (seed and actual sampled parameters), normalized intercepts/max rates, normalized training and testing points and weights, activity-noise/regularization convention, neuron model and time constants, and identity target definition controlled. Decoders are refitted in physical units. The proof requires the same activity matrix and regularizer; merely saying "same neuron count" is insufficient.

3d. A product requires both values jointly: pairs with the same sum can have different products, such as $(1, 0)$ and $(1/2, 1/2)$. A joint 2-D population preserves both coordinates for nonlinear decoding. Radius $\sqrt{2}$ covers all corners of $[-1, 1]^2$. Enlarging a radius stretches the same normalized representation across a larger physical domain; corresponding absolute identity errors stretch with it. Relative error need not worsen in that controlled comparison. The exact identity scaling proof does not directly establish the same formula for every nonlinear target.

4. LIF calibration and sweeps

4a. At fixed $J > 1$, increasing either time constant increases the interspike interval and lowers the rate. The asymptotic ceiling is $1/\tau_{ref}$. This reasoning holds current fixed; it does not describe a recalibrated maximum-rate experiment by itself.

4b. Rearrange the rate equation:

$$\ln(1 - 1/J_{max}) = \frac{\tau_{ref} - 1/a_{max}}{\tau_{RC}}, \qquad J_{max} = \frac{1}{1 - \exp[(\tau_{ref} - 1/a_{max})/\tau_{RC}]}.$$

The two constraints $\alpha z + \beta = 1$ and $\alpha + \beta = J_{max}$ give

$$\alpha = \frac{J_{max} - 1}{1 - z}, \qquad \beta = 1 - \alpha z.$$

For the supplied values, the exponent is -0.4, so $J_{max} \approx 3.033$, $\alpha \approx 2.033$, $\beta = 1$. The current at the intercept is threshold current 1, not zero current.

4c. The new exponent is -0.25, so $J_{max} \approx 4.521$, $\alpha \approx 3.521$, and $\beta = 1$. Both tuning curves are silent for $r \leq 0$, begin above threshold just to the right of zero, and reach 100 Hz at $r = 1$. Their interiors generally differ because the rate nonlinearity changes even after the endpoint currents are recalibrated. For example, at $r = 1/2$ the old rate is about 63 Hz and the new rate about 71 Hz. A higher refractory period lowers the rate at fixed current, but the new gain increases current to restore the specified endpoint.

4d. Sweep one time constant at a time, holding normalized intercepts, target maximum rates, encoders, radius, neuron realization, normalized evaluation domain, solver/noise convention, and other time constant fixed. Recompute gains and biases to preserve the tuning constraints; recompute activities and fit decoders because tuning curves changed. Sketch several curves and compare identity RMSE over the same domain, preferably across paired seeds. Static rate-based decoding accuracy is different from temporal spiking reconstruction accuracy.

Finite positive J_{max} requires $0 < a_{max} < 1/\tau_{ref}$. At 5 ms the ceiling is 200 Hz, reached only as current tends to infinity. An exact 200 Hz draw is infeasible at finite gain; rates strictly below it remain feasible, potentially with very large gains near the ceiling. Neither these constraints nor a qualitative tuning sketch guarantee monotonic population RMSE: shape, coverage, regularization, and decoder fit interact.

5. Feedforward and observed steps

5a. The ideal identity plateaus are $0, 1, 0$. Draw a source node into scalar X, a filtered identity connection $X \to Y$, and separate filtered observation branches from X and Y. The raw input has sharp edges. The X recording is rounded by its probe. Under the ideal-source assumption, the Y recording passes through the connection filter and then its probe, so it has two filtering stages and a slower onset. Neural responses add approximation and spiking effects to these schematic predictions.

5b. Let $q = 1 - e^{-\Delta/\tau}$. The second filter obeys $\tau\dot{p} + p = q$, $p(0) = 0$. Substitution verifies

$$p = 1 - (1 + \Delta/\tau)e^{-\Delta/\tau}, \qquad \tau\dot{p} + p = 1 - e^{-\Delta/\tau}.$$

At $\Delta = \tau$, the one-stage value is $1 - e^{-1} \approx 0.632$, the two-stage value $1 - 2e^{-1} \approx 0.264$. Define causal step responses

$$S_1(v) = \begin{cases} 0 & v < 0 \\ 1 - e^{-v/\tau} & v \geq 0, \end{cases} \quad S_2(v) = \begin{cases} 0 & v < 0 \\ 1 - (1 + v/\tau)e^{-v/\tau} & v \geq 0. \end{cases}$$

For $\tau = 0.010$ s, the X recording is $S_1(t - 0.1) - S_1(t - 0.4)$; the Y recording is $S_2(t - 0.1) - S_2(t - 0.4)$ under the idealizations. Differences of shifted responses include the residual tail at pulse offset.

5c. The ideal affine plateaus are $1, -1, 1$. One implementation fits a function decoder to $1 - 2x$ over the source domain, then uses an identity transform into Y. Another sends $-2x$ plus a separate constant represented input of 1. A scalar transform k computes kx and vanishes at $x = 0$; it cannot supply the intercept. The receiving neuronal biases calibrate the receiver's tuning curves, not a freely chosen common represented offset of 1. Both desired plateaus fit a scalar radius 1, with practical margin sometimes helpful.

5d. With zero-initialized filters and a constant target present from $t = 0$, the recording generally starts at zero and approaches its baseline of 1. Startup conditions matter. After initial settling, the schematic affine recording is a baseline near 1 with a negative, twice-amplitude rounded pulse and recovery to 1. A correct plateau with rounded transitions is consistent with filtering. A sustained wrong in-range plateau suggests static decoding error or a construction issue; range mismatch is another possibility. More neurons can improve representation but do not remove the prescribed linear filters.

6. Desired dynamics and recurrent transforms

6a. Substitution gives

$$\dot{\mathbf{x}} = \frac{A' - I_d}{\tau}\mathbf{x} + \frac{B'}{\tau}\mathbf{u}.$$

Therefore $A' = I_d + \tau A$ and $B' = \tau B$. Shapes are $A, A' : d \times d$ and $B, B' : d \times p$. The identity compensates the filter's leak of the state, so it acts on the state space. Adding a state identity to the input map can be dimensionally invalid. With consistent state/input units, the transformed maps produce state-valued quantities; τA is dimensionless.

6b. For $\dot{\mathbf{x}} = \mathbf{u}$, $A = 0_{d\times d}$, $B = I_d$, $A' = I_d$, $B' = \tau I_d$. At 50 ms, recurrent transform is 1 and input transform is 0.05 in the scalar model using seconds. Input transform 1 would yield $\dot{x} = u/\tau = 20u$, integrating twenty times too fast under matched filters and otherwise ideal representation. Radius affects representability, not this desired scaling.

6c. $A = -2\,\mathrm{s}^{-1}$ and $B = 1$, giving $A' = 0.9$ and $B' = 0.05$. The ideal state is $x(t) = 0.6e^{-2t}$ with time constant 0.5 s. The 0.9 is the value transform inside a continuous low-pass recurrence, not a discrete state multiplier applied once per second. The filter equation supplies the relationship between transform and derivative.

6d. For matched filters,

$$\dot{x} = \frac{-x + x + \epsilon(x) + \tau u}{\tau} = u + \frac{\epsilon(x)}{\tau}.$$

With zero input and $\epsilon = 0.001$, drift is $0.001/0.05 = 0.02$ state units per second. If $\epsilon(x_*) = 0$ and $\epsilon'(x_*) < 0$, nearby states converge to x_* locally in this scalar model. Static decoder distortion can replace a continuum of ideal held values with isolated attracting or repelling equilibria. Smoother rate dynamics can therefore still drift.

7. Definite-integral predictions

7a. The ideal state is

$$x(t) = \begin{cases} 0, & 0 \le t < 0.04, \\ 0.9(t - 0.04), & 0.04 \le t < 1, \\ 0.864, & t \ge 1. \end{cases}$$

It is continuous at both events; $0.9(1 - 0.04) = 0.864$. Thus $x(1.5) = 0.864$. The input is the slope of the state, so stopping input stops the increase rather than erasing the accumulated area. Sketch a flat start, rising line, then horizontal hold.

7b. For the short pulse,

$$x(t) = \begin{cases} 0, & 0 \le t < 0.04, \\ 0.9(t - 0.04), & 0.04 \le t < 0.16, \\ 0.108, & t \ge 0.16. \end{cases}$$

The area is $0.9(0.12) = 0.108$, one eighth of the longer pulse's area. Both are within scalar radius 1. A short pulse probes faster transitions; a smaller held value can make similar absolute distortion/noise large in relative terms. No universal better/worse neural-error ranking follows from area alone.

7c. The ramp slope is $0.9/0.45 = 2$ input units per second:

$$u(t) = \begin{cases} 2t, & 0 \le t < 0.45, \\ 0, & t \ge 0.45, \end{cases} \qquad x(t) = \begin{cases} t^2, & 0 \le t < 0.45, \\ 0.2025, & t \ge 0.45. \end{cases}$$

Triangle area is $\frac{1}{2}(0.45)(0.9) = 0.2025$. The state is a convex parabola while the input ramps; it holds after the input drops to zero. The derivative changes abruptly at 0.45 s but the state remains continuous.

7d. Evaluate both limits:

$$x(t) = \int_0^t 5 \sin(5t')\, dt' = [-\cos(5t')]_0^t = 1 - \cos(5t).$$

The period is $2\pi/5$ s and the range is $[0, 2]$. Radius 1 is insufficient for the intended state trajectory. A symmetric radius of at least 2 covers the ideal state; practical margin can accommodate deviations. The constant term 1 is fixed by $x(0) = 0$, not an arbitrary fitting parameter. The indefinite antiderivative $-\cos(5t)+C$ becomes this unique definite integral. Increasing radius can increase absolute identity decoding error under the normalized-domain controls of Stage 3; the representational choice is a coverage/accuracy tradeoff.

8. Error mechanisms and controlled feedback

8a. Rate mode usually reduces fast fluctuations caused by individual spikes. Static decoder distortion/regularization, recurrent amplification of that distortion, filter transients, observation lag, limited trained domain, and numerical discretization can remain. Three distinct examples suffice. The feedback decoder remains a finite approximation, so eliminating spikes does not eliminate $\epsilon(x)$ or guarantee a perfectly held state.

8b. The derivation assumed the input and recurrent contributions share the same filter. PN3's 5 ms and 50 ms synapses violate it; its instructed transforms use the recurrent 50 ms, giving $A' = 1, B' = 0.05$. Using $B' = 0.005$ merely because the input filter is 5 ms is not the matched-filter derivation.

Mismatch can alter transient slopes and produce overshoot/offset during changing input. In the ideal linear, unlimited representation model with these transforms, it does not by itself imply persistent zero-input drift: after all transients settle, a finite pulse still produces its intended total area. Sustained drift is consistent with approximate recurrent feedback or neural noise. A 10 ms probe smooths the recorded state but does not change the state equation.

Compare otherwise identical models with both connection synapses at 50 ms, and compare rate/spiking versions across the same several neuron seeds with identical inputs and probe settings. Consistent edge distortions suggest filtering; seed-dependent held-state behavior suggests representation/feedback effects. These comparisons inform diagnosis rather than uniquely proving a mechanism.

8c. For the explicitly defined target $\dot{x} = u + cx$,

$$g'(u) = 0.05u, \qquad f'(x, c) = x + 0.05cx.$$

With zero input and constant c, $x(t) = 0.4e^{ct}$. Thus $c = 0$ holds 0.4, $c = -2$ decays as $0.4e^{-2t}$, and $c = +2$ grows as $0.4e^{2t}$ while the ideal trajectory remains representable. With $u = 0.3, c = -2$, equilibrium is $x_* = 0.15$ and $\dot{x} = -2(x - 0.15)$ attracts nearby states. Here c has units s^{-1}; it controls feedback, rather than gating only the external input.

8d. Use scalar X and two-dimensional C. Route $X \to C$ through $(x, 0)^T$; route the normalized external control into coordinate 2 as $(0, \tilde{c})^T$. Decode

$$F(C) = C_1 + \tau c_0 C_1 C_2 = C_1 + 0.1 C_1 C_2$$

from C back into scalar X. Send $0.05u$ directly into X through the intended input filter. For $|x| \le 1, |\tilde{c}| \le 1$, radius $\sqrt{2}$ covers the joint box; scale domains appropriately if larger states/control values are intended. The extra routing stage can delay/distort the joint representation; this two-population realization approximates the algebraic feedback recipe and should be checked over its intended bandwidth. A paper demonstration: accumulate a positive pulse at $c = 0$, stop input and hold, change to $c = -2$ to predict exponential decay, then briefly use $c = +2$ to predict growth within range.

Optional. For $\dot{x} = u - x^3$, set $f'(x) = x - \tau x^3$ and $g'(u) = \tau u$. At $u = 0$, positive states decrease and negative states increase toward zero. The derivative vanishes at zero; the attraction is nonlinear and is not an exponential with a fixed nonzero linear decay rate near zero.

9. Independent readiness: compact key

9a. Four incoming transforms are $I_2, -3I_2, 2I_2, -2I_2$. The contributions sum to $(0.6, 0.5)^T$. A radius at least $\sqrt{0.61} \approx 0.781$ covers that single ideal output; a larger input trajectory requires its own norm check. The neural weight matrix is 60×80, of form $E_{post}TD_{pre}$, with radius included in E_{post}.

9b. The predicted RMSE is $0.08(0.5/2) = 0.02$, assuming matching normalized activities and samples, the same linear solver/regularizer, neuron realization, and identity targets scaled by radius. For fixed intercept z and maximum rate a_{max}, recalibrate

$$J_{max} = [1 - e^{(\tau_{ref} - 1/a_{max})/\tau_{RC}}]^{-1}, \quad \alpha = (J_{max} - 1)/(1 - z), \quad \beta = 1 - \alpha z.$$

Require $z < 1$ and $0 < a_{max} < 1/\tau_{ref}$ for this finite-gain calibration, then recompute activities and decoders. Interior tuning shapes can change; monotonic RMSE is not guaranteed.

9c. Ideal plateaus are $1, -1, 1$. Decode the entire affine function, or combine transform -2 with a separate represented constant 1. A probe on the source sees one filtering stage under the ideal-source assumption, while the receiver recording sees connection and probe filtering, so edges are rounded and delayed further. The baseline may rise from zero at startup. The receiver's neuronal bias sets individual tuning/current offsets; it is not automatically the function's represented intercept.

9d. From $\tau\dot{\mathbf{x}} = -\mathbf{x} + A'\mathbf{x} + B'\mathbf{u}$, coefficient matching gives $A' = I + \tau A, B' = \tau B$. The scalar integrator uses $A = 0, B = 1, A' = 1, B' = 0.04$. Its pulse state is

$$x(t) = \begin{cases} 0, & 0 \le t < 0.1, \\ 0.6(t - 0.1), & 0.1 \le t < 0.4, \\ 0.18, & t \ge 0.4. \end{cases}$$

Separately, $\int_0^t 2\sin(4t')dt' = \frac{1}{2}(1 - \cos 4t)$, with range $[0, 1]$.

9e. Rate drift can arise from static error in recurrent decoding; spiking adds temporal fluctuations and noise that can accumulate through feedback. Probe smoothing reduces recorded ripple and adds lag but does not repair the state. Unequal connection synapses violate the common-filter derivation and change transient behavior. In the ideal linear integrator they do not alone create indefinite drift after input/filter transients settle. More neurons can improve approximation/averaging, but prescribed lag, domain limitations, seed variability, and error tradeoffs prevent a universal per-run monotonic RMSE guarantee.

9f. Use $f'(x, c) = x + 0.04cx$ and $g'(u) = 0.04u$. Draw scalar state X routed with external scaled c into a joint 2-D population, then decode this product-containing feedback into X. Respect joint radius and added routing dynamics. At zero input, $c = 0$ holds the initial state ideally; $c = -1\,\mathrm{s}^{-1}$ gives $x(t) = x(0)e^{-t}$, with a 1 s decay time. A fixed linear map of (x, c) is a sum of constant multiples of each; its sensitivity to x cannot vary with c, whereas cx requires that dependence. Joint nonlinear tuning plus function decoding supports approximating the product.

What to revisit. Value/current or weight dimensions: Stages 1–2. Radius controls: Stage 3. Calibration and feasibility: Stage 4. Affine offsets and filters: Stage 5. Recurrence transforms or drift: Stage 6. Initial conditions and areas: Stage 7. Rate/spike comparison or variable feedback: Stage 8. Redo the corresponding readiness part independently before the practice paper.